

DEBUGGING EXAM

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Course code:CSA0307

Course name: DATA STRUCTURES

QUESTION-1:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name 'Kothamasu Srinath' with a dropdown arrow. The main content area is titled 'QUESTIONS' and lists three questions. Question 1 is selected and shows a code snippet for a linear search function. The code is as follows:

```
int linearSearch(int arr[], int n, int key) {
    for(int i = 0; i <= n; i++) {
        if(arr[i] == key)
            return i;
    }
    return -1;
}
```

Below the code, there is a 'Debugging Task' section with the instruction: 'Find and fix the bugs in the code shown above. Write your corrected code in the editor below.' The 'Your Fixed Code' section contains the following code:

```
#include <stdio.h>
int linearSearch(int arr[], int n, int key) {
    for(int i = 0; i < n; i++) {
        if(arr[i] == key)
            return i;
    }
    return -1;
}
int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = sizeof(arr) / sizeof(arr[0]);
    int key = 30;
    int result = linearSearch(arr, n, key);
    if(result != -1)
        printf("Key found at index: %d\n", result);
    else
        printf("Key not found\n");
    return 0;
}
```

On the right side, there is a 'Test Input' section with a text input field labeled 'stdin input (optional)' and an 'Output' section with a dark blue box.

QUESTION-2:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name 'Kothamasu Srinath' with a dropdown arrow. The main content area is titled 'QUESTIONS' and lists three questions. Question 2 is selected and shows a code snippet for a pop operation in a stack. The code is as follows:

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Below the code, there is a 'Debugging Task' section with the instruction: 'Find and fix the bugs in the code shown above. Write your corrected code in the editor below.' The 'Your Fixed Code' section contains the following code:

```
#include <stdio.h>
#define MAX 100
int stack[MAX];
int top = -1;
int pop() {
    if (top < 0) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}
int main() {
    printf("Popped: %d\n", pop());
    stack[++top] = 10;
    stack[++top] = 20;
    printf("Popped: %d\n", pop());
    printf("Popped: %d\n", pop());
    printf("Popped: %d\n", pop());
    printf("Popped: %d\n", pop());
    return 0;
}
```

At the bottom of the code editor, there are three buttons: 'Run', 'Save', and 'Submit Fix'. On the right side, there is a 'Test Input' section with a text input field labeled 'stdin input (optional)' and an 'Output' section with a dark blue box.

QUESTION-3:

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SIMATS Online Programming Lab
DS PROGRAMMING

Kothamasy Srinath
PRTS42200A

QUESTIONS

Q1
1. Find the bugs. Include the missing lines in the following snippet.
10 marks

Q2
6. Debug Code in underflow operation.
10 marks

Q3
11. Create a LL with 5 nodes and display even nodes.
1 mark

3/3 answered

Q3 11. Create a LL with 5 nodes and display even nodes 1 mark

```
void display(struct Node *head)
{
    while(head != NULL){
        {
            printf("%d ", head->data);
            head = head->next;
        }
    }
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7 void displayEvenNodes(struct Node *head) {
8     int pos = 1;
9     while (head != NULL) {
10         if (pos % 2 == 0) {
11             printf("%d ", head->data);
12         }
13         head = head->next;
14         pos++;
15     }
16 }
17 int main() {
18     struct Node *head = NULL, *tail = NULL;
19     for (int i = 1; i <= 5; i++) {
20         struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
21         newNode->data = i * 10;
22         newNode->next = NULL;
23         if (head == NULL) {
24             head = newNode;
25             tail = newNode;
26         } else {
27             tail->next = newNode;
```

Test Input

stdin input (optional)

Output